



Kubernetes



kubernetes.io (<https://kubernetes.io/>)

Automated container deployment, scaling, and management

Kubernetes is a cluster management system for managing containerized applications across multiple hosts, providing mechanisms for deployment, maintenance, and scaling of applications. It groups containers that make up an application into logical units for easy management and discovery.

Kubernetes builds upon years of experience at Google running production workloads at scale using a system called Borg (<https://research.google.com/pubs/pub43438.html>), combined with best-of-breed ideas and practices from the community. Google released Kubernetes in 2014, donated it as a seed technology for the founding of the Cloud Native Computing Foundation (<https://www.cncf.io/announcement/2015/06/21/new-cloud-native-computing-foundation-to-drive-alignment-among-container-technologies/>)

in 2015, and continues actively using and developing the project.

Repo

Kuberne...

License

Apache ...

Categories

Cloud...

Develop...

Containe...

Language

Go...

How Google uses Kubernetes

Kubernetes powers Google Kubernetes Engine, a part of Google Cloud Platform, upon which many new products and services are built.

Related Projects



(<https://opensource.google/projects/firebase-sdk>)

Firestore

(<https://opensource.google/projects/firebase-sdk>)

Firestore is an app development platform that provides integrated tools to help you build, grow and monetize your apps. The



(<https://opensource.google/projects/apache-beam>)

Apache Beam

(<https://opensource.google/projects/apache-beam>)

Apache Beam provides an advanced unified programming model, allowing you to implement batch and streaming data



(<https://opensource.google/projects/knative>)

Knative

(<https://opensource.google/projects/knative>)

Knative components build on top of Kubernetes, abstracting away the complex details and enabling developers to focus on what matters. Built by codifying



(<https://opensource.google/projects/istio>)

Istio

(<https://opensource.google/projects/istio>)

Istio is an open platform that provides a uniform way to connect, manage, and secure microservices. Istio supports managing traffic flows between microservices, enforcing

Firebase SDK enables access to the Firebase services in an intuitive and idiomatic manner on several platforms.

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processing jobs that can run on any execution engine. It is easy to use with Apache Apex, Apache Flink, Apache Spark, and Google Cloud Dataflow among other distributed processing back-ends.

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the best practices shared by successful real-world implementations Knative solves the “boring but difficult” parts of deploying and managing cloud native services so you don’t have to.

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access policies, and aggregating telemetry data, all without requiring changes to microservice code.

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